

Quiz for BTech Chemistry (CYL1010)

Total Marks = 20

Time = 30 mins

PART-I (10 x 1 = 10)

1. A particle is confined to a 1D box of length L . Which of the following statements about the energy levels of a particle is correct?
 - a) The energy levels are continuous ✓
 - b) The energy levels are discrete
 - c) The energy levels are determined by the width of the box *
 - d) The energy levels are independent of the box length *
2. Consider a particle in 1D box. The length of the box is halved while keeping the potential energy constant, how the energy level change?
 - a) The energy level will decrease
 - b) The energy level will increase
 - c) The energy level will remain same
 - d) The energy level will become continuous
3. The wavenumber (in cm^{-1}) corresponding to $2.5 \mu\text{m}$ is
 - a) 1400 cm^{-1}
 - b) 4000 cm^{-1} ✓ $\left(\frac{1}{\lambda}\right)$
 - c) 3600 cm^{-1}
 - d) 400 cm^{-1}
4. Which of the following transitions mainly occurs in IR?
 - a) Electronic transitions only
 - b) Rotational and vibrational transitions
 - c) Rotational transitions only
 - d) All electronic, rotational, and vibrational
5. How many vibrational modes are possible for HCl?
 - a) 0
 - b) 1
 - c) 2 ✓ $3n - 5 = 2$
 - d) 3
 - e) 1 & 2
6. Two molecules AB and B₂ have same force constant and vibrating in simple harmonic motion. The atomic weight of A is 5 times the atomic weight of B. The ratio of zero-point energy of AB and B₂ is
 - a) $\sqrt{2/3}$ ✓ $m_A = 5m_B$
 $6m_B$ $2m_B$ $\left(\frac{n+1}{2}\right)\omega$ $\sqrt{\frac{k}{m}}$
 - b) $\sqrt{3/2}$
 - c) $\sqrt{3/5}$ $\sqrt{\frac{k}{6m_B}}$ $\sqrt{\frac{k}{2m_B}}$
 - d) $\sqrt{5/3}$
7. If we change the reduced mass of a SHO (simple harmonic oscillator) from 1μ to 16μ . The fundamental vibrational frequency will
 - a) Increase by 4 times
 - b) Decrease by 4 times ✓ $\frac{\nu}{\nu_0} = \sqrt{\frac{m_0}{m}}$
 - c) Increase by 2 times
 - d) Decrease by 2 times

8. For an energy level, the J value is 6. What will be the degeneracy and energy value? (Given B = 10.0 cm^{-1})

a) 13,70

b) 7,70

c) 36,420

d) 13,420

9. Find out the values of (n, l, and m) and assign orbital to the wave function

$$\psi_{n,l,m} = \frac{1}{\sqrt{32\pi}} \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} (2-\sigma) e^{-\sigma/2}$$

a) 1,0,0

b) 2,0,0

c) 3,0,0

d) 4,0,0

10. What is the wavelength of the light emitted (in nm) when the electron in a hydrogen atom undergoes transition from the energy level with n = 4 to energy level n = 2? (Given $R_H = 109678 \text{ cm}^{-1}$)

a) 386 nm

b) 486 nm

c) 546 nm

d) 268 nm

PART-II (2 x 5 = 10)

- 1. Derive wave function for a particle in 1D box, and prove that zero-point energy for a particle in 3D box is three times of a particle in 1D box.

2. Show that average value of $1/r$ for an electron in the 1s orbital of hydrogen atom is $1/a_0$, where a_0 is the Bohr radius, given that

$$\psi_{1s} = \frac{1}{\sqrt{\pi}a_0^{3/2}} e^{-r/a_0}$$

Indian Institute of Technology Jodhpur
CYL1010
Minor Exam
Inorganic Section

1. Draw the structures of the following molecules and identify their shape. (2.5 marks)
a) TeCl_4 , b) NOCl , c) BrF_3 d) SnCl_3^- e) IF_4^-
2. Determine the point groups for the following molecules. Also indicate C_n , and σ symmetry elements present in the molecules (2.5 marks)
a) $[\text{NO}_2]^+$ b) POCl_3 c) XeF_4 d) $[\text{BF}_4]^-$ e) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
3. a) Show that the body centred cubic lattice is less economically packed compared to the face centred cubic lattice. (1.5 marks)
b) Sodium crystallises in body centred cubic structure with $a = 4.24 \text{ \AA}$. Calculate the i) theoretical density of Na and ii) the radius of Na-atom ($M_{\text{Na}} = 23$) (1.5 marks)
4. The planes in a crystalline solid intersect the crystal axes at $(2a, b, c)$, $(-a, b, c)$, $(a, 2b, 3c)$, and $(-a, b, \infty)$. Determine the Miller indices for these planes. (1 mark)
5. Pick up the correct answer. (1 mark)
a) The ratio of the nearest neighbour atomic distance in body centred (bcc) and face centered cubic (fcc) crystals with the same unit cell edge length is:
A) 1.225 B) 0.866 c) 0.707 d) 0.5
b) The point group and total number of unique symmetry operations for Al_2Cl_6 molecule will be
a) C_{2h} and 4 b) D_{2h} and 8 c) D_{2d} and 8 d) C_{2v} and 8 e) D_{2h} and 4



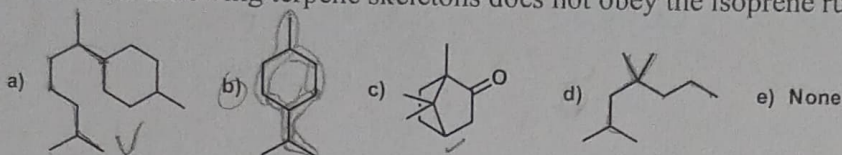
Indian Institute of Technology Jodhpur
Department of Chemistry
Semester-II (Jan-Apr-2023)
Minor, Sub: Chemistry (Organic) (CYL1010) - BTech

Time – 120 Min. (23/02/2025)

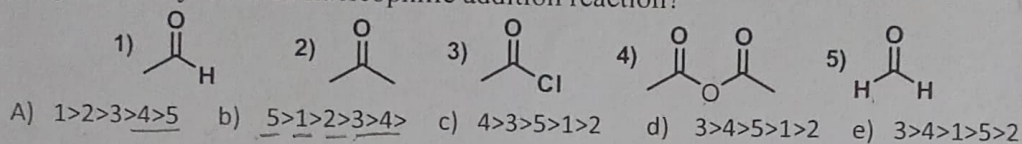
Total Marks = 20 + In-organic

Q.1. MCQ. [06 Marks]

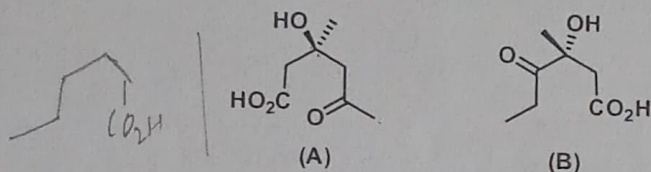
1] Which of the following terpene skeletons does not obey the isoprene rule?



2] In terms of chemoselectivity, which of the following is correct order of functional group reactivity towards nucleophilic addition reaction?

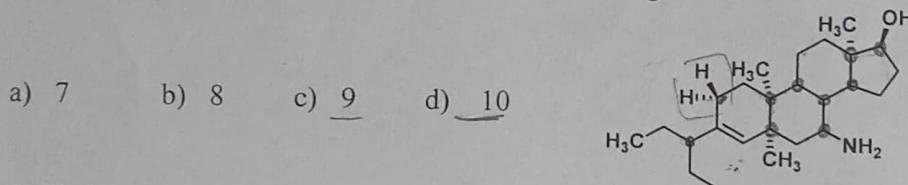


3] Relationship between (A) and (B) is

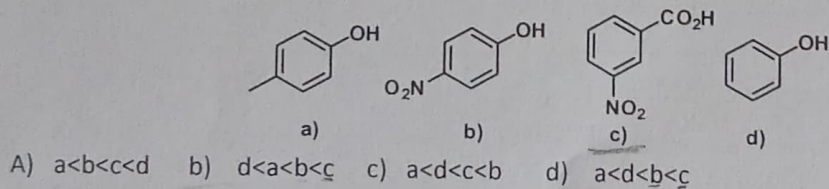


a) Enantiomer ☒ b) Identical ☒ c) Diastereomer d) Isomers

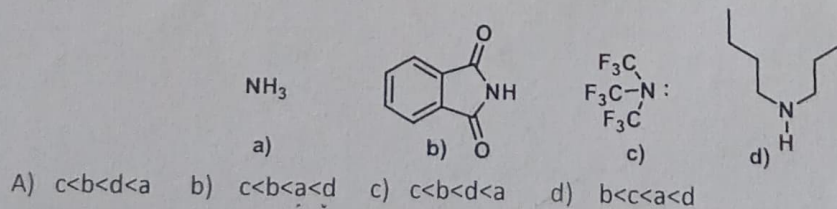
4] How many stereocenter/s presents in the following steroid molecule



5] Find the correct order of the following compounds as per their acidity.

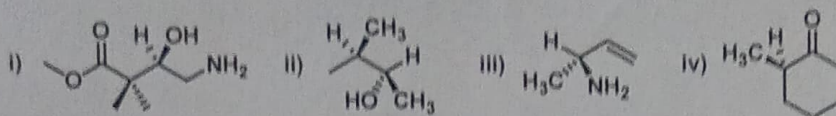


6] Find the correct order of the following compounds as per their basicity.

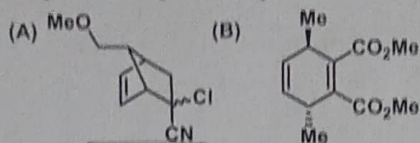


Q.2. SAQ. [14 Marks]

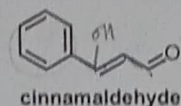
- Assign the configuration's (R/S) for each stereocenter present in the following molecules. [2 Marks]



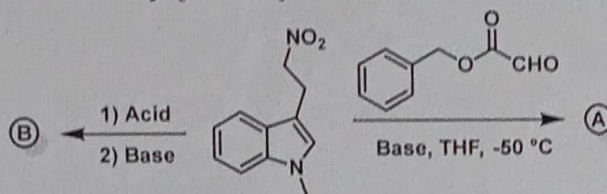
- Considering Stereospecificity & stereoselectivity, what dienes and dienophiles would react to give the following Diels-Alder products? [2 Marks]



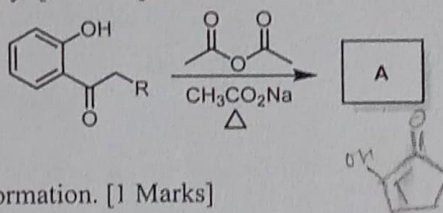
- Cinnamaldehyde is used as a flavouring agent in cinnamon candies. Show how cinnamaldehyde is synthesized by a crossed aldol condensation followed by dehydration. [2 Marks]



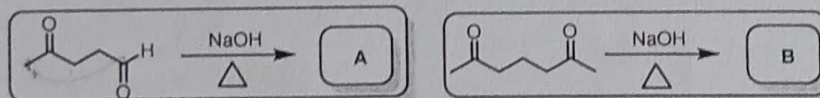
- Predict the named reaction and find the products A and B using mechanistic pathway and intermediate formation if any. [3 Marks]



- Predict the named reaction and find the product A using mechanistic pathway and intermediate formation if any. [2 Marks]



- Find the products A & B formation. [1 Marks]



- Complete the total synthesis of ibuprofen by finding the reagent B and products A, C and D. [2 Marks]

